

Technical and Executive Report: main.py (v1)

1. Executive Summary

This report summarises the performance assessment of the predictive model **main.py (v1)**. The purpose of the analysis was to evaluate the accuracy of the model when trained and tested using the supplied training dataset.

The model uses a RandomForestRegressor approach with engineered time-series features, including lag variables and rolling averages. The dataset was processed chronologically to reduce the risk of future data leakage.

The reported model performance metrics were: Mean Absolute Error (MAE): 151,662.26 Root Mean Squared Error (RMSE): 217,659.53 Mean Absolute Percentage Error (MAPE): 291.81% R^2 Score: 0.2263 The R^2 value indicates that the model explains approximately 22.63% of the variance in the target variable within the test dataset. The MAPE value is high, indicating substantial percentage-level prediction error.

The most influential features in the model were historical lag and rolling-average variables, particularly: rolling_6 rolling_3 lag_3 lag_12 lag_6 A key limitation identified is that the dataset was trimmed during preprocessing. Rows containing missing lag or rolling-window values were removed before modelling. This may reduce the representativeness of the training data and affect model generalisability.

The analysis demonstrates that the model can identify some predictive patterns in the dataset, but overall predictive performance remains limited based on the reported evaluation metrics.

2. Background and Purpose

The purpose of this analysis was to assess the predictive accuracy of the model **main.py (v1)** when trained on the supplied training dataset.

The model appears to have been designed as a time-series forecasting model using historical observations and categorical descriptors to predict future values.

The intended audience for this report includes both technical and executive stakeholders.

3. Methodology Summary

Data Preparation

The following preprocessing steps were applied: The dataset was loaded from training_dataset.csv. Date values were converted into datetime format. Numeric target values were coerced into numeric form. Records with missing date or value fields were removed. Data was sorted chronologically. **Feature**

Engineering

Additional predictive features were created, including: Year Month Lag features for 1, 2, 3, 6, and 12 periods Rolling averages for 3 and 6 periods Lag features were grouped by: Description Geography Sex Age Rows containing missing lag or rolling values were subsequently removed from the dataset.

Encoding

Categorical variables were transformed using one-hot encoding for: Description Geography Sex Age
Source file **Training and Testing Approach**

The dataset was split chronologically: 80% training data 20% testing data No shuffling was applied in order to preserve time ordering.

Model Configuration

The following Random Forest configuration was used: Number of estimators: 300 Maximum features: square root Random state: 42 Parallel processing enabled (n_jobs=-1)

4. Key Findings

The model produced the following results on the test dataset.

5. Limitations and Caveats

Trimmed Dataset

The dataset was trimmed during preprocessing because rows containing missing lag and rolling-window values were removed.

This process may: Reduce the available sample size. Exclude early observations from each grouped series. Affect representativeness of the dataset. **Model Performance Constraints**

The reported R^2 score of 0.2263 indicates relatively limited explanatory power.

The MAPE value of 291.81% suggests substantial percentage-level prediction error, which may limit operational reliability depending on intended use.

Time-Series Assumptions

[ASSUMPTION: The modelling approach assumes that temporal relationships in the historical data remain sufficiently stable for forecasting purposes.]

Missing Contextual Information

The following information was not provided and therefore could not be assessed: Dataset size Forecast horizon Business acceptance thresholds Benchmark model comparisons External validation results Hyperparameter optimisation rationale

6. Appendix

Model Run Results

Metric	Result
MAE	151,662.26
RMSE	217,659.53
MAPE	291.81%

R ²	0.2263
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Top 10 Feature Importances

Feature	Importance
rolling_6	0.131813
rolling_3	0.117988
lag_3	0.110516
lag_12	0.103388
lag_6	0.101524
description_patients waiting less than 26 weeks (all stages of pathway)	0.092985
lag_2	0.085183
lag_1	0.084742
source_file_rtt_patient_waiting_length.csv	0.074543
year	0.043922